Project 3: Implementation of the Coriolis Force in a Spherical Navier–Stokes Solver

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1 Introduction

The objective of this project is to adapt a two-dimensional Navier–Stokes solver operating on the surface of a unitary sphere manifold to incorporate a Coriolis force term, and to calibrate the resulting dynamics to Earth-like atmospheric regimes using physically meaningful non-dimensional parameters.

2 Theoretical Formulation

2.1 The Navier–Stokes Equations on a Sphere

Let us start by considering the NS equations in **velocity form** and we already add the Coriolis force term $f = 2\Omega \sin\phi\, \kappa \times \mathbf{u}$, where:

- $\mathbf{u}$ is the fluid velocity vector, tangent to the surface;
- κ is the unit vector along the rotation axis of the sphere (z-axis);
- Ω is the angular rotation rate of the sphere;
- ϕ is the latitude at the considered point on the sphere.

For the sake of simplicity we will henceforth always use $2\Omega \sin\phi = \ell$.

$$\frac{\partial \mathbf{u}}{\partial t} + (\mathbf{u} \cdot \nabla)\mathbf{u} + \ell\, \kappa \times \mathbf{u} = -\nabla p + \nu\, \Delta \mathbf{u}. \tag{1}$$

2.2 Motivation Behind the Vorticity Formulation

From this form we want to move to the **vorticity form** by applying the curl operator $\nabla\times$ to each term:

$$\nabla \times \left[\frac{\partial \mathbf{u}}{\partial t} + (\mathbf{u} \cdot \nabla)\mathbf{u} + \ell\, \kappa \times \mathbf{u}\right] = \nabla \times \left[-\nabla p + \nu\, \Delta \mathbf{u}\right]. \tag{2}$$

Using standard vector calculus identities, we note that:

- The vorticity contribution from the Coriolis force can be derived as follows. Starting from the curl of a scalar times a vector, we use the standard vector calculus identity:

  $$\nabla \times (\ell\, \mathbf{C}) = (\nabla \ell) \times \mathbf{C} + \ell\, (\nabla \times \mathbf{C}), \tag{3}$$

  where in our case $\mathbf{C} = \kappa \times \mathbf{u}$.

  For incompressible flow on the sphere (tangential velocity $\mathbf{u}$), the second term vanishes:

  $$\nabla \times (\kappa \times \mathbf{u}) = \kappa(\nabla \cdot \mathbf{u}) - (\kappa \cdot \nabla)\mathbf{u}, \tag{4}$$

because $\nabla \cdot \mathbf{u} = 0$ and there is no variation along the rotation axis in 2D tangential flow. Hence, we are left with:

$$\nabla \times (\ell\, \kappa \times \mathbf{u}) = (\nabla \ell) \times (\kappa \times \mathbf{u}). \tag{5}$$

Next, we introduce a **stream function** $\psi = \Delta^{-1}\omega$ on the sphere such that

$$\mathbf{u} = \nabla^{\perp}\psi, \tag{6}$$

where $\nabla^{\perp}\psi$ denotes the tangential rotation of the surface gradient. Solving $\psi = \Delta^{-1}\omega$ is equivalent to solving the Poisson problem studied in class.

To ensure that the velocity field is divergence-free ($\nabla \cdot \mathbf{u} = 0$) on the manifold $\mathcal{S}$ and to guarantee the uniqueness of the stream function by removing the constant kernel of the Laplace–Beltrami operator, we impose the zero-mean constraint:

$$\int_{\mathcal{S}} \psi \, dA = 0. \tag{7}$$

In the numerical implementation, we enforce this property to ensure a well-posed problem for the vector of coefficients $\underline{\Psi}$.

Finally, the curl of the Coriolis term becomes

$$\nabla \times (\ell\, \kappa \times \mathbf{u}) = \nabla^{\perp}\psi \cdot \nabla \ell \equiv J(\psi, \ell). \tag{8}$$

- The nonlinear advection term of the vorticity can be rewritten in terms of the stream function as follows. Starting from the general identity:

$$\nabla \times \big((\mathbf{u} \cdot \nabla)\mathbf{u}\big) = (\mathbf{u} \cdot \nabla)\boldsymbol{\omega} - (\boldsymbol{\omega} \cdot \nabla)\mathbf{u}, \tag{9}$$

where $\boldsymbol{\omega} = \nabla \times \mathbf{u}$ is the vorticity vector normal to the surface.

On a 2D manifold, the velocity is tangential and the vorticity points along the normal direction, so the vortex stretching term vanishes: $(\boldsymbol{\omega} \cdot \nabla)\mathbf{u} = 0$.
Hence, only the vorticity advection term remains:

$$\nabla \times \big((\mathbf{u} \cdot \nabla)\mathbf{u}\big) = (\mathbf{u} \cdot \nabla)\omega\, \kappa, \tag{10}$$

where $\omega = (\nabla \times \mathbf{u}) \cdot \kappa$ is the scalar vorticity along the normal direction κ which is the only non-zero direction in our case.

Finally, expressing the tangential velocity in terms of the **stream function**, $\mathbf{u} = \nabla^{\perp}\psi$, we obtain

$$\nabla \times \big((\mathbf{u} \cdot \nabla)\mathbf{u}\big) = \nabla^{\perp}\psi \cdot \nabla \omega \equiv J(\psi, \omega). \tag{11}$$

- $\nabla \times \frac{\partial \mathbf{u}}{\partial t} = \frac{\partial \boldsymbol{\omega}}{\partial t}$, where $\boldsymbol{\omega} = \nabla \times \mathbf{u}$ is the vorticity vector.

- $\nabla \times \nabla p = 0$, so the **pressure term vanishes**. The primary motivation for this transformation is that the curl of a gradient is identically zero. By applying the curl operator, the pressure term vanishes completely from the dynamics. This effectively decouples the velocity field from the pressure, allowing us to solve for the flow without the need to satisfy the complex Inf-Sup (LBB) stability conditions typically required for pressure-velocity coupling in the finite element method.

- $\nabla \times (\nu \Delta \mathbf{u}) = \nu \Delta \boldsymbol{\omega}$.

Dimensional Reduction: In a 2D framework (or on the surface of a sphere), the vorticity vector $\boldsymbol{\omega} = \nabla \times \mathbf{u}$ is always orthogonal to the plane of motion. Since $\mathbf{u}$ is tangent to the surface (lying in the local (x, y) plane) and we also have the divergence-free constraint, we basically go from 3 velocity degrees of freedom to just one, which can be described using the vorticity vector that simplifies to $\boldsymbol{\omega} = (0, 0, \omega_z)$. This represents a significant simplification: the problem of finding a velocity vector $\mathbf{u}$ and a scalar pressure p is reduced to solving for a single **scalar function** ω_z, which we shall call ω.

By putting together all the pieces, the dynamics of an incompressible fluid on a two-dimensional manifold with free boundary conditions are governed by the Navier–Stokes equations in the vorticity–stream function (ω, ψ) formulation:

$$\begin{cases} \partial_t \omega + J(\psi, \omega + \ell) - \nu \Delta \omega = 0, \\ \Delta \psi = \omega, \end{cases} \tag{12}$$

where $\omega = \nabla \times \mathbf{u}$ is the scalar relative vorticity, $\mathbf{u} = \nabla^\perp \psi$ is the velocity field, and by linearity, the Jacobian:

$$J(\psi, \omega + \ell) = \nabla^\perp \psi \cdot \nabla(\omega + \ell)$$

represents the total advection (transport) of absolute vorticity. In this framework, the Coriolis effect is no longer treated as an external force, but as an intrinsic geometric property of the rotating manifold that generates vorticity whenever fluid moves meridionally.

3 Finite Element Discretization

Rewriting the Transport Term

Using the vector calculus identity

$$\nabla \cdot (f\mathbf{a}) = \nabla f \cdot \mathbf{a} + f\, \nabla \cdot \mathbf{a}, \tag{13}$$

with $f = \omega + \ell$ and $\mathbf{a} = \nabla^\perp \psi$, we obtain

$$\nabla \cdot \big((\omega + \ell)\nabla^\perp \psi\big) = \nabla(\omega + \ell) \cdot \nabla^\perp \psi + (\omega + \ell) \underbrace{\nabla \cdot (\nabla^\perp \psi)}_{=0}. \tag{14}$$

Since $\nabla \cdot (\nabla^\perp \psi) = 0$, the transport term simplifies to the conservative form

$$\nabla^\perp \psi \cdot \nabla(\omega + \ell) = \nabla \cdot \big((\omega + \ell)\nabla^\perp \psi\big). \tag{15}$$

Thus the strong equation becomes

$$\partial_t \omega + \nabla \cdot \big((\omega + \ell)\nabla^\perp \psi\big) = \nu \Delta \omega + f. \tag{16}$$

3.1 Weak Formulation

Let φ be a test function. Multiplying by φ and integrating over Ω by parts (closed surface, so boundary terms vanish):

$$\int_\Omega \varphi\, \partial_t \omega\, dA - \int_\Omega (\omega + \ell)\nabla^\perp \psi \cdot \nabla \varphi\, dA - \nu \int_\Omega \nabla \varphi \cdot \nabla \omega\, dA = \int_\Omega \varphi f\, dA. \tag{17}$$

3.2 Finite Element Discretization

Let $\{\varphi_i\}_{i=1}^N$ be the $P1$ Lagrange shape functions.

We approximate

$$\omega_h(x,t) = \sum_{j=1}^N \omega_j(t)\,\varphi_j(x), \quad \psi_h(x,t) = \sum_{j=1}^N \psi_j(t)\,\varphi_j(x), \quad \ell_h(x) = \sum_{j=1}^N \ell_j(x)\,\varphi_j(x). \tag{18}$$

Define the coefficient vectors

$$\underline{\Omega} = \begin{pmatrix} \omega_1 \\ \vdots \\ \omega_N \end{pmatrix}, \qquad \underline{\Psi} = \begin{pmatrix} \psi_1 \\ \vdots \\ \psi_N \end{pmatrix}, \qquad \underline{\ell} = \begin{pmatrix} \ell_1 \\ \vdots \\ \ell_N \end{pmatrix}. \tag{19}$$

The mass matrix and stiffness matrix remain the same:

$$M_{ij} = \int_\Omega \varphi_i \varphi_j \, dA, \qquad S_{ij} = \int_\Omega \nabla\varphi_i \cdot \nabla\varphi_j \, dA. \tag{20}$$

3.3 Semi-discrete System

Testing with φ_i gives

$$M\dot{\underline{\Omega}} - T(\underline{\Omega} + \underline{\ell}, \underline{\Psi}) - \nu S \underline{\Omega} = F, \tag{21}$$

with the discrete Poisson problem

$$S\underline{\Psi} = -M\underline{\Omega}. \tag{22}$$

As we said above, in order to ensure the zero mean of our solution, we must enforce a compatibility condition on the system.We implement a set$_{z}ero_{m}eanfunctiontoshiftthesolutionPsi(onthecodetheinput$

The transport term $T(\psi, \omega + \ell)$ is integrated over the surface:

$$T(\psi, \omega+\ell)_k = \sum_{i,j=1}^N (\omega_i(t) + \ell_i)\Psi_j(t) \int_\Omega \varphi_i \, \nabla^\perp \varphi_j \cdot \nabla\varphi_k \, dA. \tag{23}$$

Because the Jacobian is bilinear, we optimize the computation by summing relative vorticity ω and planetary vorticity ℓ at each node prior to integration.

3.4 Time Integration Scheme

Let $t^m = m\Delta t$. After spatial discretization, the semi-discrete vorticity equation reads

$$M \frac{\underline{\Omega}^{m+1} - \underline{\Omega}^m}{\Delta t} + \nu S \underline{\Omega}^{m+1} = T(\underline{\Omega}^m + \underline{\ell}, \underline{\Psi}^m),$$

where M is the mass matrix, S is the Laplace–Beltrami stiffness matrix, and $T(\cdot,\cdot)$ denotes the discrete transport operator.

An implicit-explicit scheme is employed:

$$(M + \nu\Delta t\, S)\,\underline{\Omega}^{m+1} = M\,\underline{\Omega}^m + \Delta t\, T(\underline{\Omega}^m + \underline{\ell}, \underline{\Psi}^m). \tag{24}$$

The diffusion term is treated implicitly, while the transport term is handled explicitly. This choice is motivated by stability considerations.

If a fully explicit Euler scheme were used, one would obtain

$$M\underline{\Omega}^{m+1} = M\underline{\Omega}^m - \nu\Delta t\, S\underline{\Omega}^m + \Delta t\, T(\underline{\Omega}^m + \underline{\ell}, \underline{\Psi}^m),$$

which is subject to a severe parabolic stability restriction of the form $\Delta t \lesssim Ch^2/\nu$. When the spatial resolution is refined (small h, i.e. increasing the number of subdivisions), the largest eigenvalues of the stiffness matrix S grow like $O(h^{-2})$, and the explicit treatment of the viscous term may lead to numerical instabilities unless extremely small time steps are used.

By moving the diffusive contribution to the left-hand side,

$$(M + \nu\Delta t\, S)\underline{\Omega}^{m+1},$$

the stiff linear operator associated with the Laplace–Beltrami term is treated implicitly. As a consequence, the high-frequency modes (corresponding to large eigenvalues of S) are naturally damped by the inversion of the matrix $M + \nu\Delta t\, S$.

The resulting scheme achieves a balance between stability (implicit diffusion) and efficiency (explicit advection).

4 Physical Scaling and Non-Dimensionalization

4.1 Rossby Number and Dynamical Regimes

The relative importance of rotation is quantified by the Rossby number:

$$Ro = \frac{U}{fL} \sim \frac{\omega_{\max}}{2\Omega}, \tag{25}$$

where we used the scaling $U/L \sim \omega_{\max}$ and $f \sim 2\Omega$ on the unit sphere. This dimensionless parameter represents the ratio between inertial forces and Coriolis forces. The value of Ro determines the fundamental dynamical regime of the fluid:

- $Ro \gg 1$: Inertial forces dominate the dynamics, and the effects of rotation are negligible. This is exemplified in the non-rotating control case where $\Omega = 0.0$, resulting in an effectively infinite Rossby number ($Ro \approx 8.5 \times 10^6$).
- $Ro \ll 1$: The Coriolis force dominates the flow, leading to a *geostrophic* or *quasi-geostrophic* regime where the velocity field is always tangent to the contours of the stream function ψ.

In this study, a target of $Ro < 0.3$ was established to enforce an "Earth-like" atmospheric regime. With a peak vorticity $\omega_{\max} \approx 10$, a rotation rate of $\Omega \geq 20$ ensures that the simulation remains firmly within this rotationally controlled regime ($Ro \approx 0.25$), allowing for the clear identification of Rossby-wave dynamics.

Initial Calibration From the numerical initialization and the color scale in Figure 6, the peak vorticity is

$$\omega_{\max} = 8.85 \approx O(10). \tag{26}$$

With the baseline rotation rate $\Omega = 20.0$, this yields

$$Ro = \frac{10}{2 \times 20.0} \approx 0.25. \tag{27}$$

This places the simulation firmly in the rapidly rotating, Earth-like regime.

Choosing Ω for Earth-like Dynamics To enforce an Earth-like atmospheric regime with $Ro < 0.3$, the rotation rate must satisfy

$$\Omega > \frac{\omega_{\max}}{2\,Ro_{\text{target}}}. \tag{28}$$

For $Ro_{\text{target}} = 0.3$ and $\omega_{\max} = 10$, we obtain

$$\Omega > \frac{10}{0.6} \approx 16.7. \tag{29}$$

Therefore, any choice $\Omega \gtrsim 20$ guarantees Earth-like rotational control. Our choice $\Omega = 20$ provides a comfortable margin and produces clearly identifiable Rossby-wave dynamics.

4.2 Temporal Calibration

To relate dimensionless code time to physical time, we compare numerical and physical vorticity scales. Taking a characteristic atmospheric vorticity $\zeta_{\text{phys}} \approx 10^{-4}\,\text{s}^{-1}$ and $\omega_{\text{code}} = 8.85$, the time scaling is

$$\mathcal{T} = \frac{\omega_{\text{code}}}{\zeta_{\text{phys}}} = 8.85 \times 10^4\,\text{s} \approx 24.6 \text{ hours}. \tag{30}$$

Thus, one unit of code time corresponds to approximately one Earth day.

With a numerical time step $dt = 0.002$, each iteration advances

$$\Delta t_{\text{phys}} = 0.002 \times 8.85 \times 10^4 \approx 177\,\text{s} \approx 3 \text{ minutes}, \tag{31}$$

which is sufficient to resolve fast dispersive Rossby waves while maintaining stability.

5 Implementation and Simulation Details

5.1 Coriolis Parameter and Transport Logic

The Coriolis parameter is computed from the nodal z-coordinate. For efficiency, relative and planetary vorticity are combined at the element level:

```
// Inside compute_transport_coriolis
double omega_sum = OMEGA[a] + OMEGA[b] + OMEGA[c];
omega_sum += coriolis[a] + coriolis[b] + coriolis[c];
T[a] += (omega_sum * (PSI[b] - PSI[c])) / 6.0;
```

5.2 Computation and Visualization of the Velocity Field

In addition to solving the vorticity–stream function system, we implemented the explicit computation of the velocity field in order to visualize the flow dynamics directly on the surface mesh.

Recall that in the vorticity–stream function formulation, the velocity is given by

$$\mathbf{u} = \nabla^{\perp}\psi, \tag{32}$$

where on the spherical surface the perpendicular gradient is defined as

$$\nabla^{\perp}\psi = \mathbf{n} \times \nabla\psi, \tag{33}$$

with $\mathbf{n}$ the outward unit normal to the surface.

Since ψ is discretized using $P1$ Lagrange finite elements, its gradient is piecewise constant. On a triangle T with vertices a, b, c and area $|T|$, the gradient of the basis function ϕ_a is constant and directed perpendicularly to the opposite edge $\vec{e}_{bc}$ within the triangle plane:

$$\nabla \phi_a = \frac{\mathbf{n} \times (c - b)}{2|T|}. \tag{34}$$

For each triangle T, the velocity is computed from the stream function by evaluating the elementwise (constant) gradient

$$\nabla \psi\big|_T = \sum_{i \in \{a,b,c\}} \psi_i \, \nabla \phi_i.$$

The tangential velocity on T is then defined as

$$\mathbf{u}_T = \mathbf{n}_T \times \nabla \psi\big|_T,$$

where $\mathbf{n}_T$ denotes the unit outward normal to the triangle.

A continuous nodal representation is obtained via area-weighted averaging (mass lumping). For each vertex i, we set

$$\mathbf{u}_i = \frac{\displaystyle\sum_{T \ni i} \frac{|T|}{3} \, \mathbf{u}_T}{\displaystyle\sum_{T \ni i} \frac{|T|}{3}}.$$

In the figures, velocity vectors are displayed alongside the vorticity colormap to illustrate the geostrophic structure of the flow.

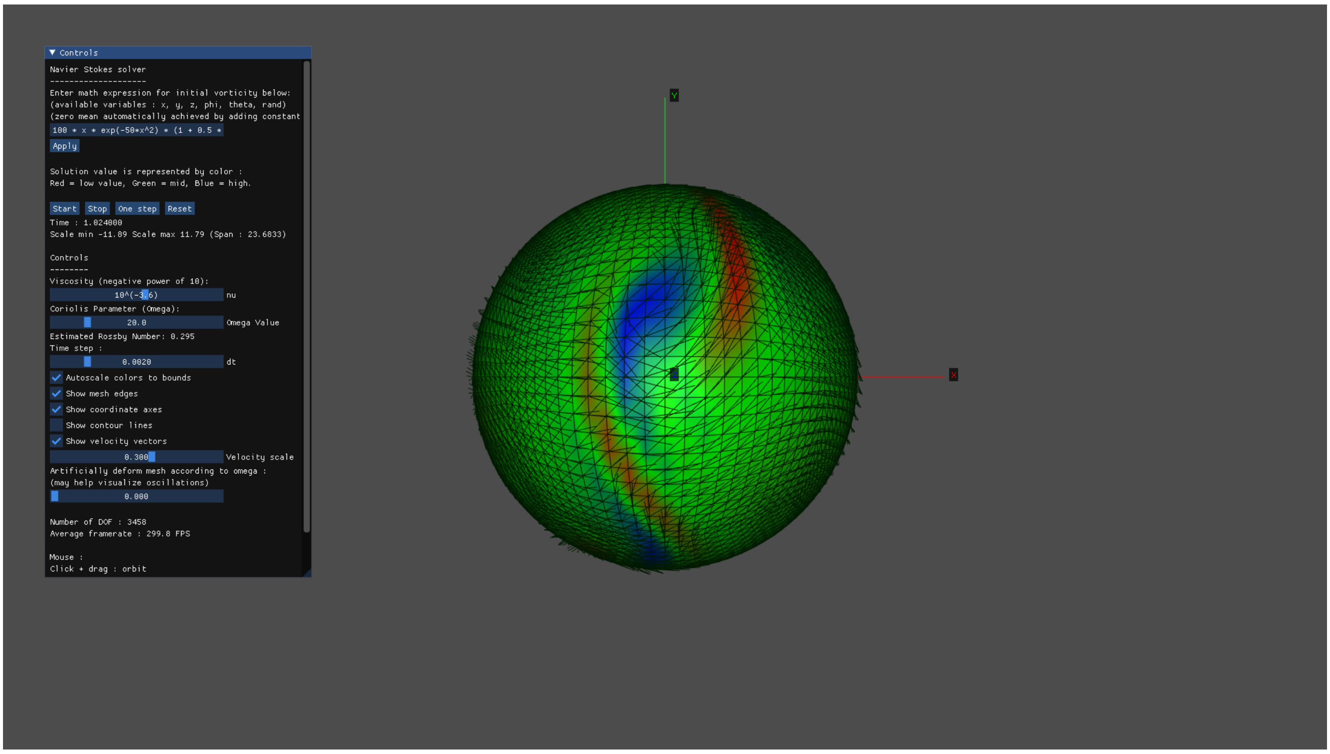

Figure 1: Vorticity field together with velocity vectors $\mathbf{u} = \nabla^{\perp}\psi$. The velocity field is tangent to the sphere and organizes around coherent vortical structures.

5.3 Streamline Visualization via Contour Plotting

To further illustrate the topological structure of the flow, we implemented a contouring algorithm to visualize the isolines of the stream function ψ.

By rendering these segments, we obtain a global view of the stream function's contours. In a rotation-dominated regime (low Ro), these contours form closed loops around vortical structures. Since $\mathbf{u} = \nabla^{\perp}\psi$, the velocity vectors are by definition tangent to these lines. The visual alignment between the white contour lines and the velocity vectors in the simulation serves as a numerical validation of the geostrophic balance.

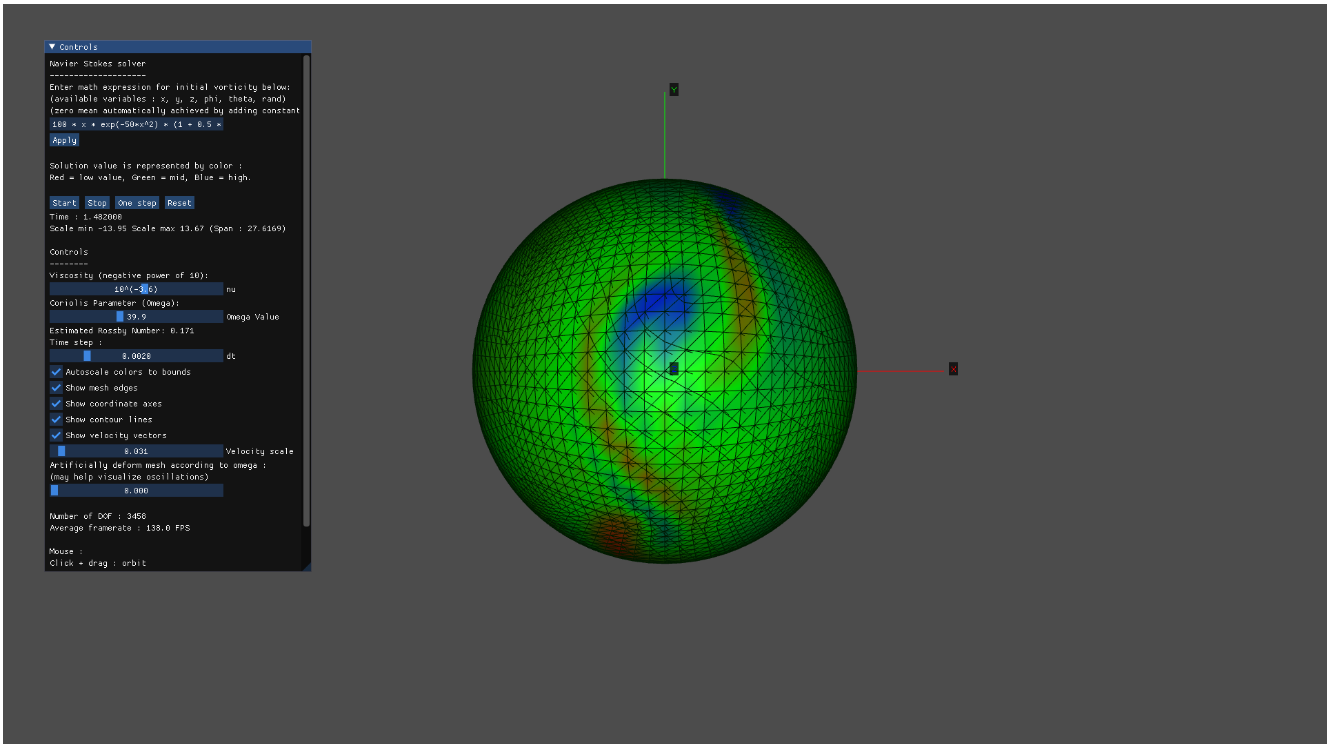

Figure 2: The velocity field is tangent to the isolines of ψ.

6 Results and Numerical Observations

Initial Condition The simulation is initialized with

$$\omega(t=0) = 100\, x\, e^{-50x^2} \left(1 + 0.5\cos\bigl(0.05\ \mathrm{atan2}(z, y)\bigr)\right), \tag{35}$$

on a spherical mesh with $n = 24$ subdivisions, corresponding to 3458 degrees of freedom. The resulting peak vorticity is $\omega_{\max} = 8.85$.

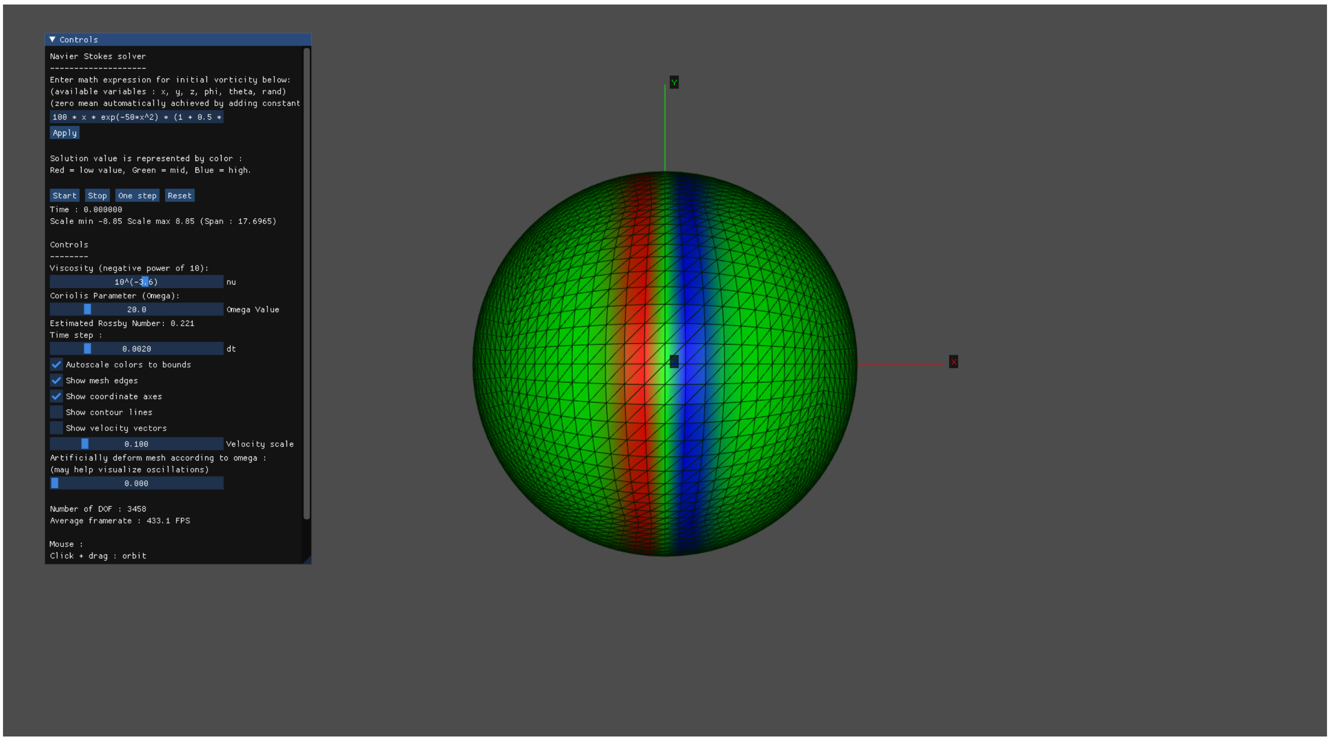

Figure 3: Initial state ($t = 0$). The vorticity is concentrated in a narrow longitudinal band.

6.1 Non-Rotating Case: Evolution and Stability without Coriolis

To establish a baseline for the effect of planetary rotation, a control simulation was performed by setting the Coriolis parameter to $\Omega = 0.0$. In this configuration, the Rossby number is effectively infinite ($Ro \approx 8.5 \times 10^6$), and the flow is governed solely by the standard incompressible Navier–Stokes equations on a spherical manifold with a viscosity of $\nu = 10^{-3.6}$.

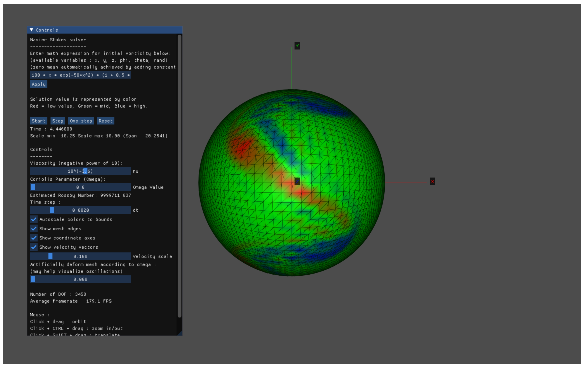

Figure 4: $t \approx 4.4$: Initial band diffusion and lack of latitudinal migration.

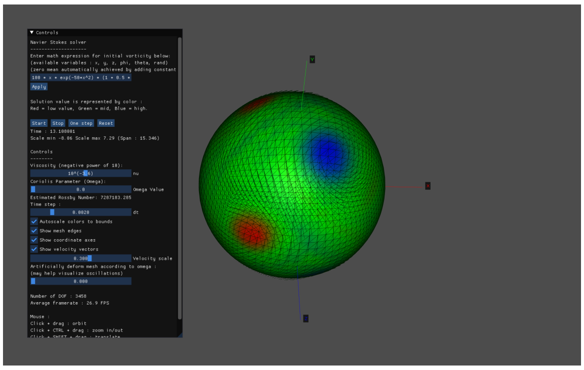

Figure 5: $t \approx 13.1$: Final state showing stable, laminar decay.

The most striking observation in this non-rotating regime is the **remarkable stability of the flow** compared to the rotating cases. While simulations with $\Omega > 0$ quickly transition into chaotic regimes and vortex roll-ups, the $\Omega = 0$ case exhibits a fundamentally different behavior.

The initial longitudinal band of vorticity does not undergo the twisting transformation: without the variation of the Coriolis parameter f, there is no mechanism to drive the fluid toward the poles. The fluid organizes itself as circular vortices that are free to move in any direction of the sphere due to the absence of latitude effect of coriolis.

The circular nature of these structures is clearly confirmed by the contour lines of the stream function where we can observe from the plot that the lines are circular close to the vortices.

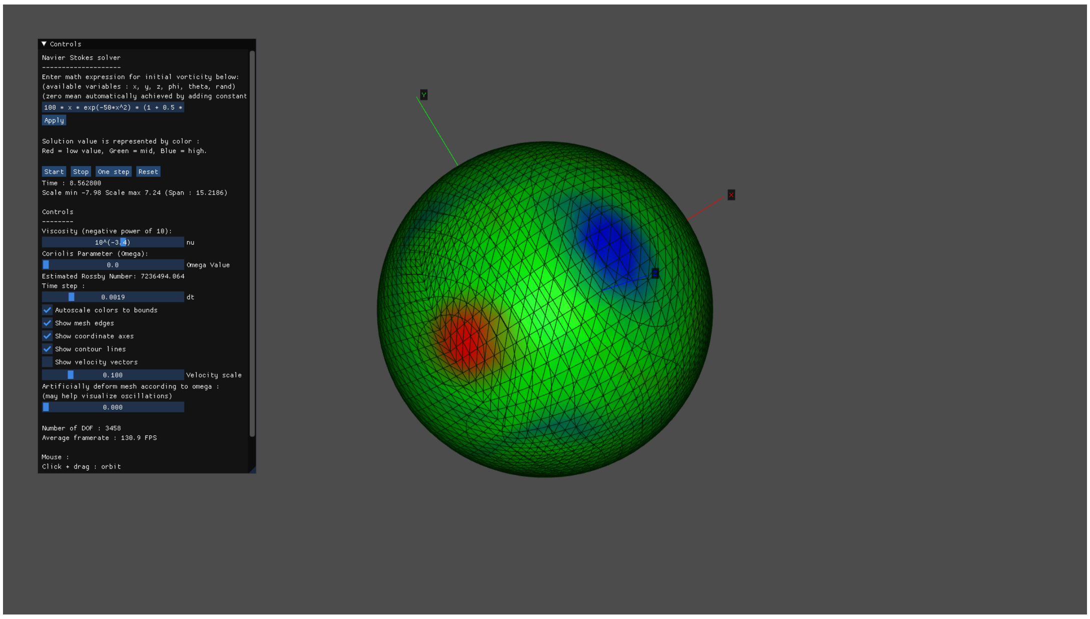

Figure 6: Countor lines of PSI are circular

The system remains perfectly stable even for $t > 13$. The absence of Coriolis-induced high-frequency oscillations and sharp gradients reduces the numerical stress on the mesh, preventing the energy accumulation at small scales that usually precedes turbulent breakdown.

6.2 Simulation with Coriolis

The transition from a non-rotating to a rotating framework ($\Omega > 0$) introduces fundamental changes to the fluid dynamics. Theoretically, we anticipate 3 primary phenomena:

1. **Latitudinal effect** When the Coriolis effect is activated, the latitudinal variation of the parameter $f = 2\Omega \sin \phi$ introduces a "twisting" effect. The initial longitudinal bands of vorticity become unstable and begin to curve.

2. **Conservation of Total Vorticity:** The quantity conserved following the motion is not the relative vorticity ω, but the absolute vorticity $\eta = \omega + f$. This conservation principle has significant dynamical implications:

   - If a fluid parcel moves meridionally toward the pole, where the planetary vorticity f increases, its relative vorticity ω must decrease (becoming more negative, i.e. anticyclonic) to compensate. The fluid is effectively driven toward the poles as a result of the conservation of η: any meridional displacement requires a sharp adjustment of the local vorticity ω, leading to the roll-up of the initial bands into coherent vortices.
   - This mechanism acts as a restoring force, enabling the existence of **Rossby waves**.These are large-scale oscillations that transport energy horizontally around the sphere. They act as an organizing mechanism, transforming chaotic fluid motion into the structured, rotation-dominated currents typical of planetary atmospheres.

3. **Polar Vortex Clustering:** Unlike the isotropic distribution of the non-rotating case, the rotating framework forces the fluid to organize into structured vortices that tend to cluster toward the poles. Since the Coriolis parameter f is maximum at the poles ($z = \pm 1$), these regions become centers of intense rotational activity.

6.2.1 Evolution Over 1 Day

Using the temporal calibration introduced above, a simulation time of

$$t \approx 1.15$$

corresponds to approximately one day of physical time. For this specific observation, the rotation rate was increased to $\Omega = 30.0$ .Given our characteristic vorticity $\omega_{\max} \approx O(10)$, the Rossby number for this run is:

$$Ro = \frac{\omega_{\max}}{2\Omega} = \frac{10.0}{60.0} \approx 0.2,$$

indicating an even more strongly rotation-dominated regime than the baseline case.

At this stage, the initial narrow longitudinal bands of vorticity have begun to destabilize. As the fluid moves meridionally (Nord-South), the latitudinal variation of the Coriolis parameter f exerts a twisting action that transforms the linear bands setting them into rotation (vortex roll-up) as they migrate toward the two poles, because when moving towards the pole the effect of coriolis increases.

These dynamics are characteristic of quasi-geostrophic flows, where the dominant Coriolis force constrains the motion.

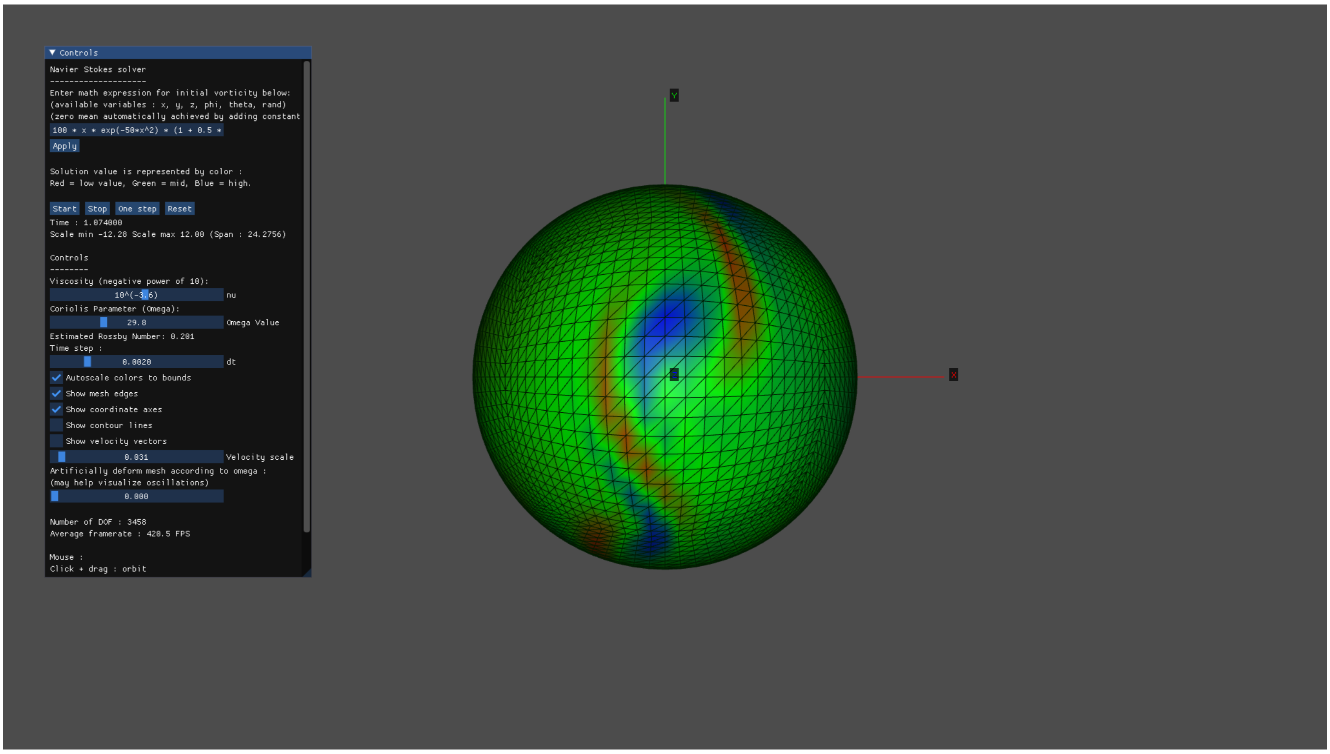

Figure 7: Snapshot at $t \approx 1$. The initial vorticity bands have destabilized and reorganized into coherent structures driven by the strong planetary rotation.

Since $Ro \approx 0.2$, the system operates in a rotation-dominated geostrophic regime. We plot the contour lines of ψ against the computed vector field to demonstrate that the velocity is indeed tangent to these lines, confirming the physical consistency of the formulation.

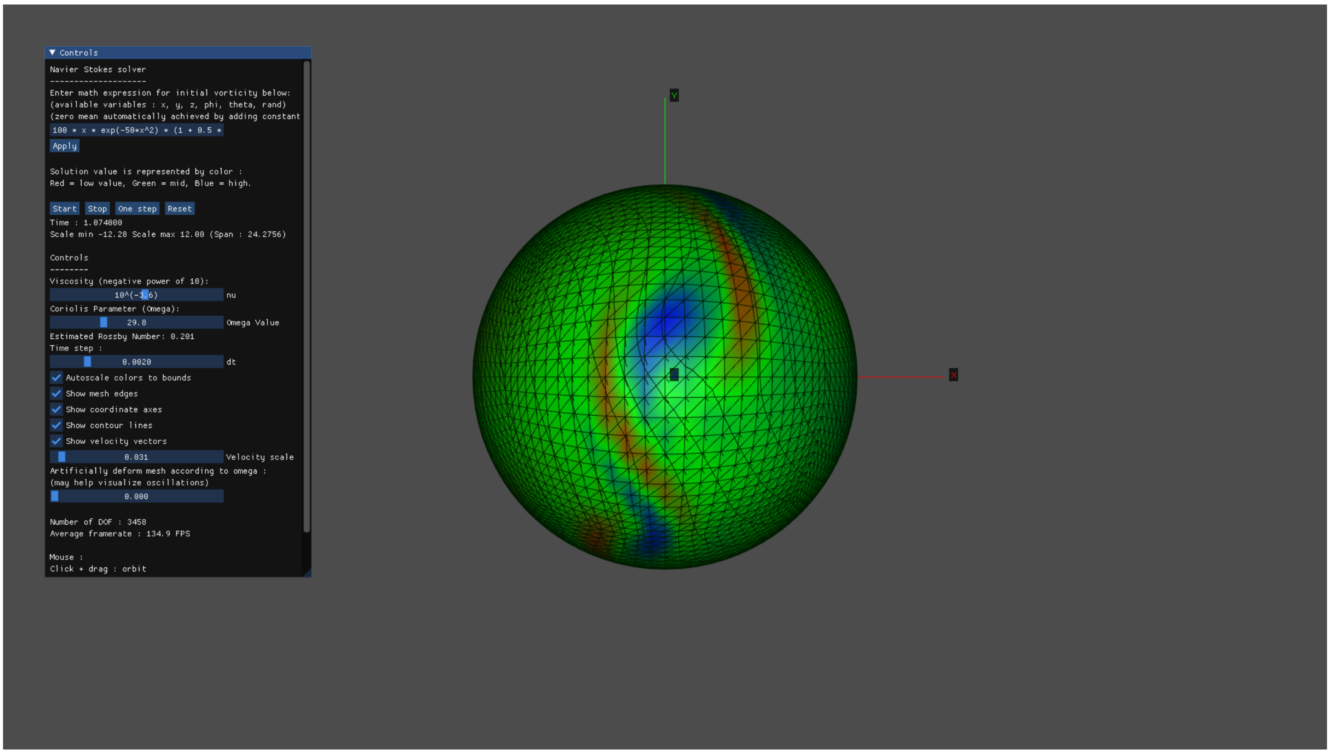

Figure 8: Snapshot at $t \approx 1$. Overlay of the velocity vector field and the stream function (ψ) contour lines, highlighting the geostrophic alignment.

6.2.2 Top View (z-axis) at $t \approx 2$ Days

Figure 9 shows a zenithal (top-down) view of the vorticity field at simulation time

$$t \approx 2,$$

corresponding to approximately two days of physical evolution. At this stage, the flow has fully transitioned into a rotation-dominated regime with

$$Ro \approx 0.3.$$

This perspective along the z-axis is particularly significant because the Coriolis parameter $f(z) = 2\Omega z$ reaches its maximum magnitude at the poles, where $z = \pm 1$. This intensified rotation at high latitudes acts as a powerful organizing mechanism for the fluid. Viewing the simulation along the z-axis (polar view) confirms this concentration of vortices at the pole.

The visualization confirms that the flow is dominated by coherent large-scale vortices close to the poles, which are typical of quasi-geostrophic dynamics.

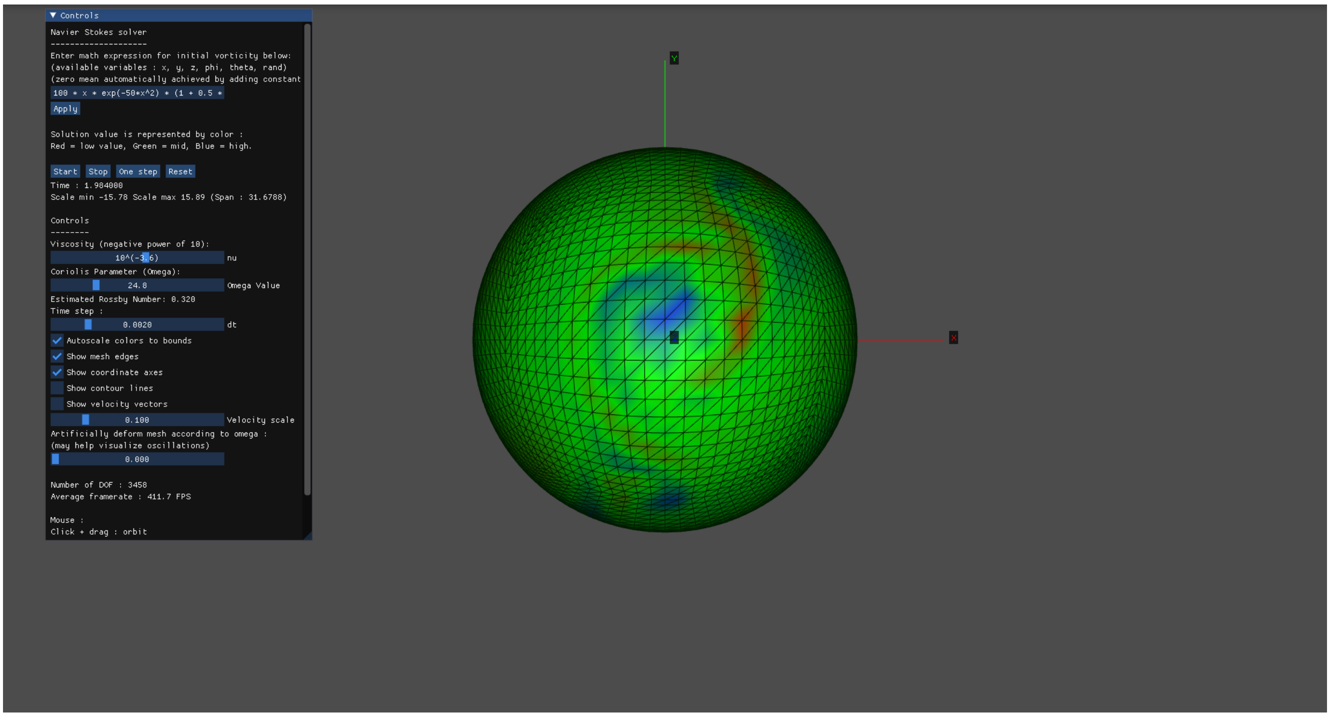

Figure 9: Top view (along the z-axis) of the vorticity field at $t \approx 2$ days. The flow is dominated by coherent large-scale vortices, characteristic of quasi-geostrophic dynamics.

The stream function ψ provides a fundamental geometric description of the flow field, where its contour lines represent the instantaneous trajectories of the fluid parcels.

The activation of the Coriolis effect dramatically alters the topology of the streamlines. As seen in the polar view (top-down along the z-axis), the contour lines are no longer scattered but tend to **cluster and nest** toward the poles.

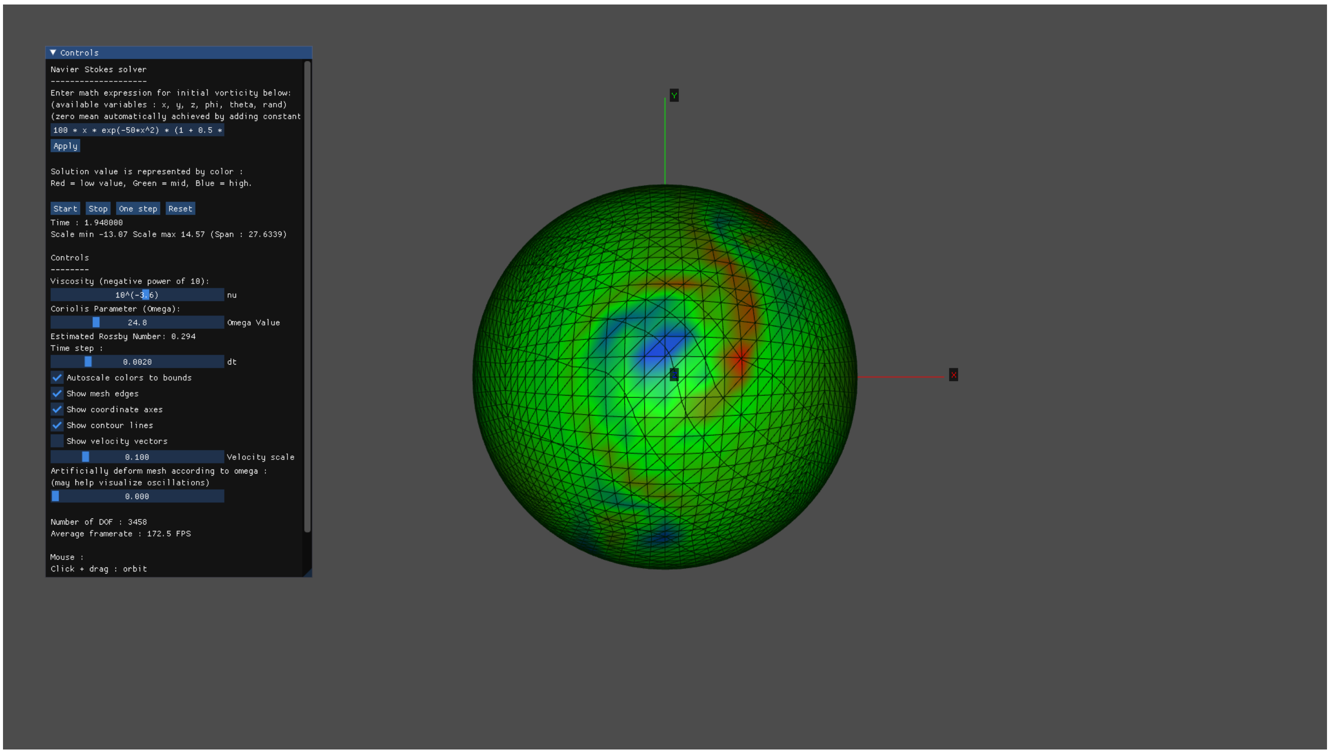

Figure 10: Top view (along the z-axis) countor lines

6.2.3 Bottom View (z-axis) at $t \approx 2$ Days

The observation of the southern pole ($-z$ axis) reveals a dynamic structure that is specularly symmetric to the northern pole, confirming the influence of spherical symmetry and the conservation of potential vorticity. At time $t \approx 2$, the system has progressed past the initial zonal band instability phase, condensing energy into a coherent polar vortex.

In contrast to the North Pole, where vorticity reaches maximum positive values (visualized in blue), the bottom view shows a central core of negative vorticity (visualized in red). This sign inversion is consistent with the definition of the Coriolis parameter $f = 2\Omega \sin(\phi)$, which becomes negative in the Southern Hemisphere ($\phi < 0$) confirming the expectations.

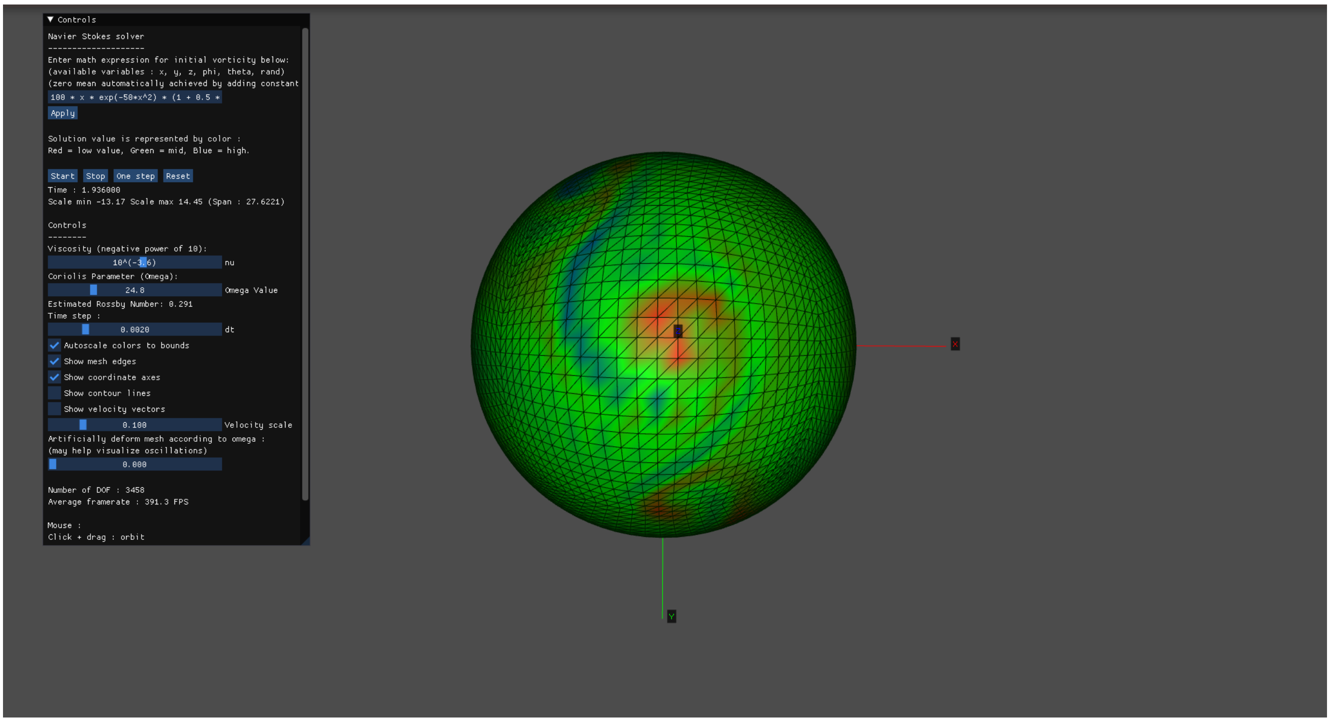

Figure 11: Bottom view (along the $-z$-axis) of the vorticity field at $t \approx 2$ days.

7 Conclusion

In this project, we implemented and analyzed a two-dimensional Navier–Stokes solver on the sphere including the effect of planetary rotation through a Coriolis term. By working in the vorticity–stream function formulation and advecting the absolute vorticity $\eta = \omega + f$, the solver successfully reproduces key dynamical features of rotating geophysical flows.

For Earth-like values of the rotation rate, corresponding to a Rossby number $Ro \lesssim 0.3$, the numerical solution exhibits a strongly rotation-dominated behavior. After an initial transient characterized by Rossby wave activity and jet destabilization, the flow rapidly organizes into large-scale coherent vortical structures on the poles.

This quasi-stationary regime persists up to approximately $t \simeq 6$ units of simulation time. Beyond this point, the solution progressively loses stability and small-scale fluctuations begin to grow. Such behavior is consistent with expectations for turbulent flows on a 2D manifold. On the other hand, the solution without Coriolis remains stable even for longer periods of simulation.